

Assignment 3 of MATP4820

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Problem 1

Consider the function

$$f(x_1, x_2) = \frac{1}{2} (2x_1 - x_2^2)^2.$$

1. Give the gradient of f at the point $(1, -1)$;

First getting $\nabla f(x_1, x_2)$:

$$\nabla f(x_1, x_2) = \begin{bmatrix} 2(2x_1 - x_2^2) \\ -2x_2(2x_1 - x_2^2) \end{bmatrix} = \begin{bmatrix} 4x_1 - 2x_2^2 \\ -4x_1x_2 + 2x_2^3 \end{bmatrix}$$

Now computing $\nabla f(1, -1)$:

$$\nabla f(1, -1) = \begin{bmatrix} 4(1) - 2(-1)^2 \\ -4(1)(-1) + 2(-1)^3 \end{bmatrix} = \begin{bmatrix} 4 - 2 \\ 4 - 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Thus, $\nabla f(1, -1) = (2, 2)$

2. Show that $\mathbf{p} = (-2, 1)$ is a descent direction;

To show $p = (-2, 1)$ is a descent direction we will show $\langle p, \nabla f(x_1, x_2) \rangle < 0$, for $(x_1, x_2) = (1, -1)$.

$$\langle p, \nabla f(x_1, x_2) \rangle = \langle p, \nabla f(1, -1) \rangle = \langle (-2, 1), (2, 2) \rangle = -2 < 0$$

Thus, $\langle p, \nabla f(x_1, x_2) \rangle < 0$, and p is a descent direction.

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3. Let $c_1 = 0.25$. Does $\alpha = 1$ satisfy the Armijo's condition at $(1, -1)$ with a search direction $\mathbf{p} = (-2, 1)$?

First computing $f(x + \alpha p)$, with $x = (1, -1)$, $\alpha = 1$ and $p = (-2, 1)$:

$$\text{With } x + \alpha p = (1, -1) + 1(-2, 1) = (-1, 0)$$

$$f(x + \alpha p) = f(-1, 0) = \frac{1}{2}(2(-1) - 0^2)^2 = \frac{1}{2}(-2)^2 = 2$$

Now computing $f(x)$, with $x = (1, -1)$:

$$f(x) = f(1, -1) = \frac{1}{2}(2(1) - (-1)^2)^2 = \frac{1}{2}(2 - 1)^2 = \frac{1}{2}$$

Now computing $\langle p, \nabla f(x) \rangle$ with $p = (-2, 1)$, $x = (1, -1)$:

$$\langle (-2, 1), \nabla f(1, -1) \rangle = \langle (-2, 1), (4(1) - 2(-1)^2, -4(-1)(-1) + 2(-1)^3) \rangle = \langle (-2, 1), (2, 2) \rangle = -2$$

Finally, computing $f(x) + c_1 \alpha \langle p, \nabla f(x) \rangle$ with $x = (1, -1)$, $c_1 = 0.25$, $\alpha = 1$, $p = (-2, 1)$:

$$f(x) + c_1 \alpha \langle p, \nabla f(x) \rangle = f(1, -1) + 0.25(1) \langle (-2, 1), \nabla f(1, -1) \rangle = \frac{1}{2} + 0.25(1)(-2) = 0$$

Now, with $f(x + \alpha p) = 2$ and $f(x) + c_1 \alpha \langle p, \nabla f(x) \rangle = 0$,

$$f(x + \alpha p) > f(x) + c_1 \alpha \langle p, \nabla f(x) \rangle$$

So Armijo's condition is not satisfied

Problem 2

Consider the function: $f(x_1, x_2) = x_1^2 - 2x_1x_2 + \frac{1}{2}x_2^2 - 4x_1 + x_2$. At the point $\mathbf{x} = (1, 0)$,

1. give the search direction vector $\mathbf{p}$ for the steepest descent method.

First, computing $\nabla f(x_1, x_2)$:

$$\nabla f(x_1, x_2) = \begin{bmatrix} 2x_1 - 2x_2 - 4 \\ -2x_1 + x_2 + 1 \end{bmatrix}$$

Now take $p = -\nabla f(1, 0)$:

$$-\nabla f(1, 0) = - \begin{bmatrix} 2(1) - 2(0) - 4 \\ -2(1) + 0 + 1 \end{bmatrix} = - \begin{bmatrix} 2 - 4 \\ -2 + 1 \end{bmatrix} = - \begin{bmatrix} -2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Thus, $p = (2, 1)$

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2. Let $c_1 = 0.25$, determine $\bar{\alpha} > 0$ such that the Armijo's condition holds for any $\alpha \in (0, \bar{\alpha}]$.

First note $\nabla f(1, 0) = (-2, -1), p = \nabla f(1, 0) = (2, 1)$

Computing $f(x + \alpha p)$:

$$\begin{aligned} f(1 + 2\alpha, 0 + \alpha) &= (1 + 2\alpha)^2 - 2(1 + 2\alpha)(\alpha) + \frac{1}{2}\alpha^2 - 4(1 + 2\alpha) + \alpha \\ &= 4\alpha^2 + 4\alpha + 1 - 2\alpha - 4\alpha^2 + \frac{1}{2}\alpha^2 - 4 - 8\alpha + \alpha = \frac{1}{2}\alpha^2 - 5\alpha - 3 \end{aligned}$$

Computing $f(x)$:

$$f(1, 0) = (1)^2 - 2(1)(0) + \frac{1}{2}(0)^2 - 4(1) + 0 = -3$$

Computing $\langle p, \nabla f(x) \rangle$:

$$\langle (2, 1), (-2, -1) \rangle = -5$$

Computing $f(x) + c_1\alpha\langle p, \nabla f(x) \rangle$:

$$-3 + 0.25\alpha(-5) = -3 - 1.25\alpha$$

Then,

$$f(x + \alpha p) \leq f(x) + c_1\alpha\langle p, \nabla f(x) \rangle$$

is satisfied when

$$\frac{1}{2}\alpha^2 - 5\alpha - 3 \leq -3 - 1.25\alpha \implies \frac{1}{2}\alpha^2 - 3.75\alpha \leq 0 \implies \frac{1}{2}\alpha \leq 3.75 \implies \alpha \leq 7.5$$

Thus, Armijo's condition holds for any $\alpha \in (0, 7.5]$

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3. Let $c_1 = 0.25$ and $c_2 = 0.5$, determine the values of $\underline{\alpha}$ and $\bar{\alpha}$ such that the Wolfe's conditions hold for any $\alpha \in [\underline{\alpha}, \bar{\alpha}]$.

We must find α such that:

$$\begin{cases} f(x + \alpha p) \leq f(x) + c_1 \alpha \langle p, \nabla f(x) \rangle \\ \langle \nabla f(x + \alpha p), p \rangle \geq c_2 \langle p, \nabla f(x) \rangle \end{cases}$$

From the previous part, $f(x + \alpha p) \leq f(x) + c_1 \alpha \langle p, \nabla f(x) \rangle$ when $\alpha \in (0, 7.5]$, so $\bar{\alpha} = 7.5$.

Now finding α such that

$$\langle \nabla f(x + \alpha p), p \rangle \geq c_2 \langle p, \nabla f(x) \rangle$$

Computing $\nabla f(x + \alpha p)$:

$$\nabla f(1 + 2\alpha, \alpha) = \begin{bmatrix} 2(1 + 2\alpha) - 2\alpha - 4 \\ -2(1 + 2\alpha) + \alpha + 1 \end{bmatrix} = \begin{bmatrix} 2\alpha - 2 \\ -3\alpha - 1 \end{bmatrix}$$

Computing $\langle \nabla f(x + \alpha p), p \rangle$:

$$\langle (2\alpha - 2, -3\alpha - 1), (2, 1) \rangle = 4\alpha - 4 - 3\alpha - 1 = \alpha - 5$$

Computing $c_2 \langle p, \nabla f(x) \rangle$

$$0.5 \langle (2, 1), (-2, -1) \rangle = 0.5(-5) = -2.5$$

Thus,

$$\langle \nabla f(x + \alpha p), p \rangle \geq c_2 \langle p, \nabla f(x) \rangle$$

holds when

$$\alpha - 5 \geq -2.5 \implies \alpha \geq 2.5 \text{ and } \underline{\alpha} = 2.5$$

So, Wolfe's condition holds when $\alpha \in [2.5, 7.5]$ and $\underline{\alpha} = 2.5$, $\bar{\alpha} = 7.5$.

Problem 3

Consider the unconstrained quadratic minimization problem:

$$\underset{\mathbf{x}}{\text{minimize}} f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x} \quad (\text{QuadMin})$$

where $\mathbf{A} \in \mathbb{R}^{n \times n}$ is a symmetric positive definite matrix, and $\mathbf{b} \in \mathbb{R}^n$.

1. Let $\mathbf{A} = \begin{bmatrix} 4 & -1 \\ -1 & 1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Set the initial vector $\mathbf{x}^0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Do by hand two iterations of steepest gradient descent with exact line search.

First Iteration:

$$\begin{aligned} p^{(0)} &= \begin{bmatrix} 2 \\ 1 \end{bmatrix} - \begin{bmatrix} 4 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ \alpha_0 &= \frac{\|p^{(0)}\|_2^2}{(p^{(0)})^T A p^{(0)}} = \frac{2^2 + 1^2}{\begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}} = \frac{5}{\begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 7 \\ -1 \end{bmatrix}} = \frac{5}{13} \\ x^{(1)} &= x^{(0)} + \alpha_0 p^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \frac{5}{13} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{10}{13} \\ \frac{5}{13} \end{bmatrix} \end{aligned}$$

Second Iteration:

$$\begin{aligned} p^{(1)} &= \begin{bmatrix} 2 \\ 1 \end{bmatrix} - \begin{bmatrix} 4 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} \frac{10}{13} \\ \frac{5}{13} \end{bmatrix} = \begin{bmatrix} -\frac{9}{13} \\ \frac{18}{13} \end{bmatrix} \\ \alpha_1 &= \frac{\|p^{(1)}\|_2^2}{(p^{(1)})^T A p^{(1)}} = \frac{\frac{81}{169} + \frac{324}{169}}{\begin{bmatrix} -\frac{9}{13} & \frac{18}{13} \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} -\frac{9}{13} \\ \frac{18}{13} \end{bmatrix}} = \frac{\frac{405}{169}}{\begin{bmatrix} -\frac{9}{13} & \frac{18}{13} \end{bmatrix} \begin{bmatrix} -\frac{54}{13} \\ \frac{27}{13} \end{bmatrix}} = \frac{\frac{405}{169}}{\frac{972}{169}} = \frac{5}{12} \\ x^{(2)} &= x^{(1)} + \alpha_1 p^{(1)} = \begin{bmatrix} \frac{10}{13} \\ \frac{5}{13} \end{bmatrix} + \frac{5}{12} \begin{bmatrix} -\frac{9}{13} \\ \frac{18}{13} \end{bmatrix} = \begin{bmatrix} \frac{75}{156} \\ \frac{150}{156} \end{bmatrix} = \begin{bmatrix} \frac{25}{52} \\ \frac{25}{26} \end{bmatrix} \end{aligned}$$

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2. Use the instructor's provided file `quadMin_gd.m` to write a Matlab function `quadMin_gd` with input **A**, **b**, initial vector **x0**, and tolerance **tol**, and with stopping condition $\|\nabla f(\mathbf{x}^k)\| \leq \text{tol}$. Also test your function by running the provided test file `test_gd.m` and compare to the instructor's function. Include your code and the results you get in your homework. **[Do your best to optimize the efficiency of your code. You will lose 10% marks if your solver takes more than double of time of the instructor's solver, and lose 20% if your solver takes more than triple of the instructor's solver.]**

Steepest Gradient Descent for Quadratic Minimization

```
function [x, hist_res] = quadMin_gd(A,b,x0,tol)

% steepest gradient method for solving
% min_x 0.5*x'*A*x - b'*x

% get the size of the problem
n = length(b);

x = x0;

% compute gradient of the objective
grad = b - A * x;

% evaluate the norm of gradient
res = norm(grad);

% save the value of res, i.e., the norm of grad
hist_res = res;

while res > tol
    % compute the stepsize alpha by exact line search
    alpha = res^2 / (grad'*A*grad);

    % update x
    x = x + alpha * grad;
```

```
% compute gradient of the objective
grad = b - A * x;

% evaluate the norm of gradient
res = norm(grad);

% save the value of res
hist_res = [hist_res; res];
end

end
```

Output of test_gd.m (Comparison to instructor's function)

```
>> test_gd
Results by student code
Total running time is 0.0270
Final objective value is -4972.1422
Results by Instructor code
Total running time is 0.0271
Final objective value is -4972.1422
```

Problem 4

Let $\mathbf{A} = \begin{bmatrix} 4 & -1 \\ -1 & 1 \end{bmatrix}$.

1. Given $\mathbf{p}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$, find a vector $\mathbf{p}_2$ with unit 2-norm, i.e., $\|\mathbf{p}_2\|_2 = 1$, and $\mathbf{A}$ -conjugate to $\mathbf{p}_1$.

Let $p_2 = \begin{bmatrix} x \\ y \end{bmatrix}$ for some x, y . First, for p_2 to be $\mathbf{A}$ -conjugate to p_1 ,

$$0 = \langle p_2, \mathbf{A}p_1 \rangle = \langle p_2, \begin{bmatrix} 4 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} \rangle = \langle p_2, \begin{bmatrix} 6 \\ -3 \end{bmatrix} \rangle = 6x - 3y$$

Now for p_2 to be unit 2-norm, $\|p_2\|_2 = 1$,

$$\sqrt{x^2 + y^2} = 1 \implies x^2 + y^2 = 1$$

Taking both conditions:

$$\begin{cases} 6x - 3y = 0 \\ x^2 + y^2 = 1 \end{cases} \implies y = 2x \implies x^2 + (2x)^2 = 1 \implies 5x^2 = 1 \implies x = \pm \frac{1}{\sqrt{5}}$$

Then,

$$\begin{aligned} x = \frac{1}{\sqrt{5}} &\implies y = \frac{2}{\sqrt{5}} \\ x = -\frac{1}{\sqrt{5}} &\implies y = -\frac{2}{\sqrt{5}} \end{aligned}$$

Thus, $p_2 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}$ or $p_2 = \begin{bmatrix} -\frac{1}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} \end{bmatrix}$

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2. Give two vectors $\mathbf{p}_1$ and $\mathbf{p}_2$ with unit 2-norm such that they are $\mathbf{A}$ -conjugate and also orthogonal to each other.

First, if p_1 and p_2 are unit 2-norm, they are orthonormal, and we can define

$$p_1 = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \text{ and } p_2 = \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix} \quad (90^\circ \text{ rotation} \implies \text{orthogonal}).$$

Now, we can solve

$$\begin{aligned} 0 = \langle p_1, Ap_2 \rangle &= \left\langle \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}, \begin{bmatrix} 4 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix} \right\rangle = \left\langle \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}, \begin{bmatrix} -4 \sin \theta - \cos \theta \\ \sin \theta + \cos \theta \end{bmatrix} \right\rangle \\ &= -4 \cos \theta \sin \theta - \cos^2 \theta + \sin^2 \theta + \cos \theta \sin \theta = \sin^2 \theta - \cos^2 \theta - 3 \cos \theta \sin \theta = 0 \end{aligned}$$

Now using trig properties

$$\begin{aligned} &(\sin^2 \theta - \cos^2 \theta) - 3 \cos \theta \sin \theta = 0 \\ \implies &-(\cos^2 \theta - \sin^2 \theta) - 3 \cos \theta \sin \theta = 0 \\ \implies &-\cos(2\theta) - \frac{6 \cos \theta \sin \theta}{2} = 0 \\ \implies &-\cos(2\theta) - \frac{3 \sin(2\theta)}{2} = 0 \\ \implies &-\cos(2\theta) = \frac{3 \sin(2\theta)}{2} \\ \implies &-\frac{\cos(2\theta)}{\sin(2\theta)} = \frac{3}{2} \implies \frac{\sin(2\theta)}{\cos(2\theta)} = -\frac{2}{3} \\ \implies &\tan(2\theta) = -\frac{2}{3} \implies 2\theta = \tan^{-1}\left(-\frac{2}{3}\right) \implies \theta \approx -16.85^\circ \end{aligned}$$

Now plugging in θ

$$\begin{aligned} p_1 &= \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \approx \begin{bmatrix} \cos(-16.85) \\ \sin(-16.85) \end{bmatrix} \approx \begin{bmatrix} 0.957 \\ -0.290 \end{bmatrix} \\ p_2 &= \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix} \approx \begin{bmatrix} -\sin(-16.85) \\ \cos(-16.85) \end{bmatrix} \approx \begin{bmatrix} 0.290 \\ 0.957 \end{bmatrix} \end{aligned}$$

Thus, we get orthonormal, A-Conjugate vectors

$$p_1 \approx \begin{bmatrix} 0.957 \\ -0.290 \end{bmatrix} \text{ and } p_2 \approx \begin{bmatrix} 0.290 \\ 0.957 \end{bmatrix}$$